Theoretical Analysis of a Symmetric Two-Stage Change Detector for SAR Images

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Abstract*— A procedure is developed to generate theoretical receiver operating characteristic (ROC) curves for the two-stage change detector proposed by Cha *et al.*, in order to compare its detection performance with other detectors. A modification, however, to the first stage of the two-stage detector is proposed, which introduces a symmetry that significantly simplifies analysis. The probability density function (pdf) for this symmetric two-stage detector is derived, and using this pdf, an approach for determining thresholds is proposed. The symmetric two-stage detector introduces a new parameter that effectively weights each statistic making up its two individual stages. Detection performance is explored versus this new parameter. The two-stage detector is shown to yield overall better detection performance than the sample variance ratio, Berger's alternative coherence estimator, and the classical coherence estimator. When applied to real synthetic aperture radar (SAR) data, the two-stage detector and Berger's estimator yield binary change images with more easily visible areas of change. Finally, we derive an expression for the pdf of the optimal log-likelihood ratio statistic that is simpler than that given by Preiss *et al.

***Index Terms*— Coherent change detection (CCD), likelihood ratio test (LRT), receiver operating characteristics (ROCs), synthetic aperture radar (SAR), two-stage change detector.**

I. Introduction

SYNTHETIC aperture radar (SAR) is the concept of an airborne or space-based radar generating high-resolution images of the scene underneath, by processing the complex-valued radar return signals from the scene. If we have two SAR return datasets of the same landscape, but taken at different times, an important application that arises is the detection of regions where the underlying scene has undergone a change. Change detection can be done by processing the two SAR datasets and generating an image where a black pixel denotes change and a white pixel denotes no change. This processing is often done by two methods: coherent change detection (CCD) and non-CCD (NCCD). CCD decides that there is a change between two pixels if the estimate of the coherence (magnitude of the complex correlation coefficient) is less than some threshold, which is determined by a given false alarm rate. The standard estimator that is used is the classical coherent change detector (classical CCD) [1], [2]. The ability of this detector to detect subtle changes in the scene has been demonstrated in [3]. One of the most simple and popular detectors falling under NCCD is the sample variance ratio (SVR), which tests for equality between the variances of the two pixels that are being compared [4], [5]. A general form of this detector has been introduced in [6] for polarimetric SAR, where the radar return for each pixel is a vector in general, and thus, the problem of testing equality of variances of the two pixels changes to testing the equality of two covariance matrices. The SVR generally performs better at detecting large-scale changes, such as the displacement of a vehicle, while classical CCD shows a better performance at detecting small-scale changes in the scene, such as vehicle tracks [7]. Combined coherent and NCCD techniques have been introduced in [7] and [8]. More recently, a number of change detection methods utilizing deep learning architectures have been proposed. Specifically, different methods have been introduced that use convolutional neural networks [9], [10], stacked autoencoders [11], [12], and a combination of both [13].

The focus of this article is the two-stage change detection method given in [7]. This method combines the coherent and noncoherent approaches in the form of a 2-D test statistic in order to detect change accurately at any scale. The first stage incorporates the SVR detector, and the second stage uses the alternative coherence estimator given by Berger in [14]. Change detection using a similar form of this alternative coherence estimator has been studied in [15]. The two-stage detector determines that a location has undergone a change if either one of the stages detects a change. In this article, we propose replacing the SVR with a symmetric SVR (SSVR), the probability density function (pdf) for which is derived in [16], in the context of edge detection in SAR images. This detector has been used by many authors, including Preiss *et al.* [17] and Oliver and Quegan [18], for the purpose of change detection as well, but assuming the true value of the coherence to be zero under all scenarios. However, in general, the coherence can take any nonzero value between zero and one [19]. Thus, the pdf and the cumulative distribution function (cdf) for the SSVR test statistic have been derived in this article for this general case of nonzero coherence. In addition, we also derive the joint pdf between the SSVR and the alternative coherence estimator. Using this symmetric form of the two-stage detector that uses the SSVR instead of the SVR significantly simplifies the implementation and the process of analyzing its detection performance.

In [17], the theoretical receiver operating characteristic (ROC) performance of the SSVR and the classical CCD methods have been discussed. Unlike these two detectors, the two-stage detector has a complicated form, and thus,

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computing its theoretical ROC performance is not straightforward. A heuristic approach to determine the thresholds has been proposed in [7] for which ROC curves were generated empirically, which suggests that the two-stage detector gives a better performance compared to the SVR and the classical CCD methods. In this article, a procedure to compute the theoretical ROC curves for the modified two-stage detector has been proposed, and a comparison of its ROC performance with the SSVR, classical CCD, and the alternative coherence estimator has been presented. All of these four detectors have also been applied to a real SAR dataset. Both theoretical and experimental results suggest that the two-stage detector yields an overall better detection performance compared to the other three detectors.

The rest of this article is organized as follows. In Section II, an overview of the different change detection methods is covered. A simplified expression for the pdf for the log-likelihood ratio statistic is derived in III. The symmetric two-stage detector is introduced in Section IV. A procedure to select the thresholds and generating the theoretical ROC curves is presented in Section V. Section VI explores the behavior of the symmetric two-stage detector in detail. Section VII includes a comparison of the theoretical ROC performance of all the different detectors. All these detectors are applied to a SAR dataset, and their change detection performance is compared in Section VIII. Section IX concludes this article.

II. Change Detection in SAR Images

The data model used for change detection in the present analysis is the same as the one followed in [7]. The goal is to determine whether there is a change between the two pixels f and g, which corresponds to the radar returns from the same location but taken at different times. These pixels are modeled as complex Gaussian random variables, and we use the statistical properties of the complex 2×1 random vector $\mathbf{x} = [f,\ g]^T$ in order to characterize change. $\mathbf{x}$ follows a circularly symmetric zero-mean bivariate complex Gaussian distribution with covariance matrix given by

$$\mathbf{Q} = \begin{bmatrix} \sigma_f^2 & \rho\sigma_f\sigma_g \\ \rho^*\sigma_f\sigma_g & \sigma_g^2 \end{bmatrix}, \quad \text{where } \rho = \frac{\mathbb{E}(fg^*)}{\sqrt{\mathbb{E}(|f|^2)\mathbb{E}(|g|^2)}} \tag{1}$$

is the complex correlation coefficient between f and g, and its magnitude $|\rho|$ is known as the coherence. $\mathbb{E}(\cdot)$ is the expectation operator, and $\sigma_f^2 = \mathbb{E}(|f|^2)$ and $\sigma_g^2 = \mathbb{E}(|g|^2)$ denote the variances of f and g, respectively. In practical scenarios, the coherence is generally expressed as a product of terms arising due to different physical contributions [20]. Some of these major contributors are the low clutter-to-noise ratio (CNR), imperfect matching in the acquisition techniques for primary and repeat-pass images, imperfections in the image processing techniques, and the decorrelation caused due to actual changes in the scene between the two collections [17]. If the collection and the processing techniques are properly designed, it is possible to mitigate their effects so that the decorrelation due to scene changes becomes a dominant contributor to the coherence $|\rho|$. For the purpose of change detection, a spatial window of size N around $\mathbf{x}$ is considered, which consists of pixel pairs $\mathbf{x}_k = [f_k, g_k]^T$, $k = 1, \ldots, N$, where each of these $\mathbf{x}_k$'s is assumed to be independent of each other and identically distributed as $\mathbf{x}$, i.e., $\mathbf{x}_k \sim \mathcal{CN}(\mathbf{0}_2, \mathbf{Q})$.[1] Based on the values of σ_f^2, σ_g^2, and ρ under the change and no-change scenarios, different hypothesis tests have been formulated where the null hypothesis H_0 represents no change and the alternative hypothesis H_1 represents a change between the pixels. The goal of each of these tests is to come up with a change detector, which can accurately determine whether the pixel pair under consideration has undergone a change or not. This detector consists of a function (test statistic) of the data matrix $\mathbf{X} = [\mathbf{x}_1, \ldots, \mathbf{x}_N]$. The range (known as the decision space) of this test statistic is partitioned into two sets Z_0 and Z_1, where Z_i is the region where the detector decides that H_i is true. In order to characterize the performance of any detector, two quantities are generally used: the probability of false alarm (P_{FA}) and the probability of detection (P_{D}). These probabilities are given by

$$P_{\text{FA}} = \Pr\{t(\mathbf{X}) \in Z_1;\, H_0\} = \int_{Z_1} p(t;\, H_0)dt \tag{2}$$

$$P_{\text{D}} = \Pr\{t(\mathbf{X}) \in Z_1;\, H_1\} = \int_{Z_1} p(t;\, H_1)dt \tag{3}$$

where $p(t; H_0)$ and $p(t; H_1)$ denote the pdf of the test statistic $t(\mathbf{X})$ under H_0 and H_1, respectively. Generally P_{FA} is a given constraint, and based on this, the region Z_1 is determined using (2). Once Z_1 is known, P_{D} can be computed from (3). The plot of P_{D} as a function of P_{FA} is known as a ROC curve [27] and is used to compare the detection performance of different detectors. A number of detectors have been proposed for the purpose of change detection; some of these are summarized next.

A. Sample Variance Ratio

One way of formulating a hypothesis test for change detection is to compare the variances of f and g by forming the ratio $R = \sigma_f^2/\sigma_g^2 \in (0, \infty)$. For a given pair of pixels $[f,\ g]^T$, if the value of R is close to one, i.e., when the variances σ_f^2 and σ_g^2 are approximately equal, then this pair is labeled as "no change," whereas, if R deviates further from unity, the detector declares "change." The hypothesis testing problem for this approach can be written as follows:

$$\begin{aligned} H_0 &: R \approx 1 \\ H_1 &: R \neq 1. \end{aligned} \tag{4}$$

The test statistic used for this hypothesis test is the SVR ($\hat{R}$), given by

$$\hat{R} = \frac{\hat{\sigma}_f^2}{\hat{\sigma}_g^2} = \frac{\sum_{k=1}^{N} |f_k|^2}{\sum_{k=1}^{N} |g_k|^2} \tag{5}$$

where $\hat{\sigma}_f^2 = (1/N)\sum_{k=1}^{N} |f_k|^2$ and $\hat{\sigma}_g^2 = (1/N)\sum_{k=1}^{N} |g_k|^2$ are the sample variances. An important point worth noting is

[1] $\mathbf{0}_2$ represents the zero vector of size 2×1.

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that, even though these sample variances are proportional to a complex[2] chi-squared random variable with N degrees of freedom (see [27] and [33, Appendix A]), their ratio $\hat{R}$ is not proportional to the usual complex central F random variable because f_k and g_k are not uncorrelated; in general, they have a nonzero correlation coefficient ρ. Thus, $\hat{R}$ can be thought of as a *generalized complex central F statistic* with equal degrees of freedom and, only in the case when ρ becomes zero, is $\hat{R}$ proportional to the usual complex central F statistic. The pdf for $\hat{R}$ is given by[3] [21]

$$p_{\hat{R}}(x;\, R, |\rho|, N) = \frac{\Gamma(2N)}{\Gamma(N)^2} \frac{(1-|\rho|^2)^N (x+R) R^N x^{N-1}}{[(x+R)^2 - 4xR|\rho|^2]^{N+\frac{1}{2}}} \quad (6)$$

for $0 < x < \infty$, where $\Gamma(\cdot)$ denotes the gamma function. $\hat{R}$ can be interpreted as a special case of the random variable $\tilde{F}_{\tilde{\Delta}}$ given in [22, Appendix A], where $\tilde{F}_{\tilde{\Delta}}$ is the ratio of the diagonal elements of a 2×2 complex Wishart distributed matrix. The cdf for $\tilde{F}_{\tilde{\Delta}}$ was also derived in [22], which we can use to derive the expression for the cdf of $\hat{R}$ by replacing the terms $\mathbf{v}(\boldsymbol{\theta_a})$, $\mathbf{v}(\boldsymbol{\theta_b})$, $\mathbf{R}$, L, N, $||\mathbf{a}||^2$, $||\mathbf{b}||^2$, and $\mathbf{a}^H\mathbf{b}$ in the expression for $\Pr(\tilde{F}_{\tilde{\Delta}} \leq \eta)$ in [22] by $\mathbf{e}_1$, $\mathbf{e}_2$, $\mathbf{Q}$, N, 2, σ_f^2, σ_g^2, and $\rho\sigma_f\sigma_g$, respectively, in terms of this article.[4] The final expression for the cdf of $\hat{R}$ is given by

$$\Pr\{\hat{R} \leq \eta\} = F_{\hat{R}}(\eta;\, R, |\rho|, N) = 0.5\{1 - \text{sign}[\lambda_1(\eta)]\} + \text{sign}[\lambda_1(\eta)] F_G(l(\eta), N) \quad (7)$$

for $\eta \geq 0$. In the above expression, G is a complex central F random variable with both degrees of freedom equal to N, i.e., G is the ratio of two independent complex chi-squared random variables each having N degrees of freedom. $F_G(.,N)$ denotes the cdf of G, which is given by [33]

$$F_G(x, N) = \frac{x^N}{(1+x)^{2N-1}} \sum_{k=0}^{N-1} \binom{2N-1}{k+N} x^k. \quad (8)$$

The remaining terms in (7) are defined in Appendix I, where it is also shown that the cdf reduces to a simpler form given by

$$F_{\hat{R}}(\eta;\, R, |\rho|, N) = F_G(l(\eta), N). \quad (9)$$

$\hat{R}$ decides that a pixel pair has not undergone change if it is near unity, i.e., $\eta_1 < \hat{R} < \eta_2$ such that $\eta_1 < 1 < \eta_2$; else, the detector declares change. In other words, the region Z_1 where this detector declares change is of the form $Z_1 = \{x \in \mathbb{R}_+ | x \leq \eta_1 \text{ or } x \geq \eta_2\}$ for some η_1 and η_2 such that $\eta_1 < 1 < \eta_2$.

B. Coherence Estimators

It is evident from (5) that $\hat{R}$ utilizes only the magnitude of pixels f_k and g_k. Thus, the SVR is incapable of detecting changes that occur in the phases of f_k and g_k. In particular, if the magnitude of the pixel remains unchanged, and the change only occurs in the phase, then the SVR cannot recognize such changes. Thus, in order to detect change on such a finer scale, CCD techniques are used, which utilize the phase information as well. The hypothesis test for CCD is as follows:

$$H_0 : |\rho| \approx 1$$
$$H_1 : |\rho| \lesssim 1 \quad (10)$$

which says that no change has occurred if the coherence between the two pixels is approximately near one, and a change has occurred if the coherence is approximately less than one. Depending on the landscape and the SAR system design, the actual value of the coherence under the two hypotheses can vary, the value under H_0 can go as low as 0.5, and the value under H_1 (which should ideally be zero, depicting uncorrelated pixels) can go up to 0.2 [19]. The analysis that is followed in this work allows us to consider any desired value to study the theoretical performance of change detection algorithms. Two coherence estimators can be used as test statistics for this test. The first one is the classical coherence estimator $|\hat{\rho}_c|$, given by

$$|\hat{\rho}_c| = \frac{\left|\sum_{k=1}^{N} f_k g_k^*\right|}{\sqrt{\sum_{k=1}^{N} |f_k|^2 \sum_{k=1}^{N} |g_k|^2}}. \quad (11)$$

The pdf of $|\hat{\rho}_c|$ is independent of σ_f^2 and σ_g^2 and is given by [23]

$$p_{|\hat{\rho}_c|}(x;\, |\rho|, N) = 2(N-1)\left(1-|\rho|^2\right)^N x \left(1-x^2\right)^{N-2} \cdot {}_2F_1(N, N; 1; x^2|\rho|^2) \quad (12)$$

for $x \in [0, 1)$, where ${}_2F_1(\cdot, \cdot; \cdot; \cdot)$ is the Gauss hypergeometric function [24]. The geometric mean in the denominator of the expression of ρ in (1) can be replaced by the arithmetic mean when the two variances are equal. Thus, we can get another expression for the coherence estimate when the two variances σ_f^2 and σ_g^2 are equal. This alternative coherence estimator, proposed by Berger [14], is given by

$$|\hat{\rho}_a| = \frac{\left|\sum_{k=1}^{N} f_k g_k^*\right|}{\frac{1}{2}\left(\sum_{k=1}^{N} |f_k|^2 + \sum_{k=1}^{N} |g_k|^2\right)}. \quad (13)$$

The estimator $|\hat{\rho}_a|$ provides accurate estimates of $|\rho|$ only when $\sigma_f^2 \approx \sigma_g^2$, i.e., when $R \approx 1$ [7]. The pdf of $|\hat{\rho}_a|$ under the assumption $R = 1$ is given by [7]

$$p_{|\hat{\rho}_a|}(x;\, |\rho|, N) = (2N-1)\left(1-|\rho|^2\right)^N x \left(1-x^2\right)^{N-\frac{3}{2}} \cdot {}_2F_1\left(N, N+\frac{1}{2}; 1; x^2|\rho|^2\right) \quad (14)$$

for $x \in [0, 1)$. Closed-form expressions for (12) and (14) have been presented in [25] and [26], which can simplify calculations. A detailed comparison of estimation characteristics between $|\hat{\rho}_c|$ and $|\hat{\rho}_a|$ has been shown in [7]. The region Z_1 for both of these test statistics is of the form

[2]The term complex does not indicate that the random variable itself is complex, but rather it is used to indicate that this real-valued chi-squared random variable is a result of summing magnitude squared complex Gaussians.

[3]The subscript indicates the random variable to which the pdf belongs, and variable x denotes the value of at which the pdf is being computed.

[4]$\mathbf{e}_1 = [1, 0]^T$ and $\mathbf{e}_2 = [0, 1]^T$.

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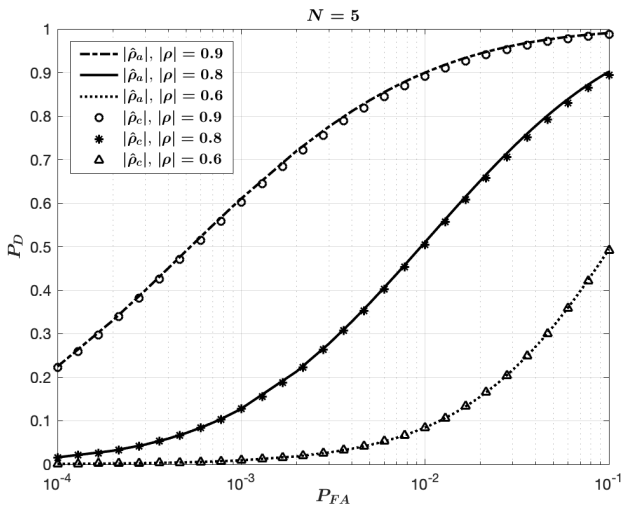


Fig. 1. ROC curves for $|\hat{\rho}_c|$ and $|\hat{\rho}_a|$ generated using different values of $|\rho|$ under H_0 (mentioned in the legend) used to compute thresholds. The change between the two pixels for all the curves is given by $|\rho| = 0$ under H_1. Change occurs due to $|\rho|$ only, i.e., $R = 1$ under both hypotheses.

$Z_1 = \{x \in \mathbb{R}_+ : x \leq \eta\}$ for some $0 < \eta < 1$. ROC curves for $|\hat{\rho}_a|$ and $|\hat{\rho}_c|$ are plotted in Fig. 1, assuming $R = 1$ under both hypotheses. Note that we can use the pdf in (14) to generate ROC curves for $|\hat{\rho}_a|$ only because we have taken $R = 1$ under both hypotheses, and in general, this pdf cannot be used if $R \neq 1$. As we can see in Fig. 1, $|\hat{\rho}_c|$ and $|\hat{\rho}_a|$ give almost the same detection performance when $R = 1$ under H_0 and H_1.

C. Two-Stage Change Detector

As mentioned earlier, NCCD methods detect changes on a larger scale, whereas CCD approaches detect change on a finer scale. The two-stage change detector was introduced by Cha *et al.* [7], which combines the alternative coherence estimator and the SVR statistics in order to detect changes with higher accuracy at all scales. The first stage of the two-stage detector is $\hat{R}$, which tests for the equality of the two variances. If the variances are not equal, it decides that a change has occurred. However, if $\sigma_f^2 \approx \sigma_g^2$, then those pixel pairs are passed on to the second stage, which is $|\hat{\rho}_a|$. Since the pixel pairs passed on to the second stage have almost equal variances, $|\hat{\rho}_a|$ provides an accurate estimation of $|\rho|$, and thus, changes can be detected on a smaller scale as well. Based on the hypotheses testing formulations for R and $|\rho|$, the change detection problem for the two-stage change detector has been formulated as a hypothesis test as follows:

$$\begin{aligned} H_0 &: R \approx 1 \text{ and } |\rho| \approx 1 \\ H_1 &: R \neq 1 \text{ or } |\rho| \lesssim 1. \end{aligned} \tag{15}$$

The two-stage change detector works as a 2-D detector, i.e., the decision space is 2-D, where the dimensions correspond to the $\hat{R}$ and $|\hat{\rho}_a|$ statistics. In order to evaluate the performance of this 2-D detector, the analysis of the joint pdf of $\hat{R}$ and $|\hat{\rho}_a|$ is required. The joint density function for the two test statistics is derived in [7] and is given by

$$\begin{aligned} &p_{|\hat{\rho}_a|,\hat{R}}(x, y; R, |\rho|, N) \\ &= \frac{(1-|\rho|^2)^N \Gamma(2N)}{\Gamma(N)\Gamma(N-1)} \frac{x}{2(y+1)^2} \\ &\quad \cdot \left[\frac{y}{(y+1)^2} - \frac{x^2}{4}\right]^{N-2} \\ &\quad \cdot \left[x|\rho| + \frac{y+R}{(y+1)\sqrt{R}}\right]^{-2N} \\ &\quad \cdot {}_2F_1\left(\frac{1}{2}, 2N; 1; \frac{2x|\rho|}{x|\rho| + \frac{y+R}{(y+1)\sqrt{R}}}\right) \end{aligned} \tag{16}$$

for $0 \leq y < \infty$, $0 \leq x \leq [4y/(y+1)^2]^{1/2}$, and the pdf is zero elsewhere. The pdf depends on the true variance ratio R, the true coherence $|\rho|$, and the spatial window size N. The joint pdf is plotted for $R = 1/2$ and $R = 2$, with $|\rho| = 0$ and $N = 5$ in Fig. 2(a) and (b).

Considering the hypothesis test in (15) and the forms of region Z_0 for the $|\hat{\rho}_a|$ and $\hat{R}$ statistics, we can see that the partition of the decision space for the two-stage detector has the form shown in Fig. 2(c), given by $Z_0 = \{(|\hat{\rho}_a|, \hat{R}) \in [0, 1) \times \mathbb{R}_+ : |\hat{\rho}_a| > \eta_3 \text{ and } \eta_1 < \hat{R} < \eta_2\}$, i.e., the detector decides that there is no change only if there is no change due to both $\hat{R}$ and $|\hat{\rho}_a|$. Throughout this article, the values $R = 1$ and $|\rho| = 0.9$ under H_0 will be used. For the H_1 case, we will be interested in two types of change scenarios: change due to $|\rho|$ only ($R = 1$ under H_1) and change due to both R and $|\rho|$ ($R \neq 1$ under H_1). The value of $|\rho|$ will be set to zero under H_1.

III. Log-Likelihood Ratio Statistic

In a binary hypothesis problem, the optimal detection performance is achieved by the log-likelihood ratio test (LRT) statistic, given that the parameters governing the two hypotheses are known [27]. Even though such knowledge is rare in real applications, the upper bound achieved by the log-likelihood ratio gives an idea about how close other realizable detectors are getting to the optimal performance. The pdf for the log-likelihood ratio statistic for the change detection problem was derived in [17], and ROC curves for this statistic were also shown. For the general case, it was shown in [17] that the log-likelihood ratio statistic reduces to a linear combination of dependent complex chi-squared random variables. However, the expression for this pdf and P_D in [17] contains infinite sums and special functions, making it time-consuming for software in terms of computation [28]. In this work, we derive a simpler expression for the pdf and the detection performance for the log-LRT statistic.

We use the same data model that has been used in [17], where it was shown that the log likelihood ratio is given by

$$z = \text{Tr}\left\{\mathbf{Q}_d \sum_{k=1}^{N} \mathbf{x}_k \mathbf{x}_k^H\right\} \tag{17}$$

where $\mathbf{Q}_d = \mathbf{Q}_0^{-1} - \mathbf{Q}_1^{-1}$, and $\mathbf{Q}_0$ and $\mathbf{Q}_1$ are the 2×2 covariance matrices of $\mathbf{x}$ under H_0 and H_1, respectively, as given in [17]. If we consider $\mathbf{Q}$ to be the true covariance of $\mathbf{x}$, we can write $\mathbf{x}_k$ in terms of a complex standard Gaussian vector as[5] $\mathbf{x}_k \stackrel{d}{=} \mathbf{Q}^{1/2}\mathbf{y}_k$ such that $\mathbf{y}_k \sim \mathcal{CN}(\mathbf{0}, \mathbf{I}_2)$, where $\mathbf{I}_2$ is

[5]The notation $\stackrel{d}{=}$ indicates equality in distribution. If two random variables A and B have the same pdf, it is denoted by $A \stackrel{d}{=} B$.

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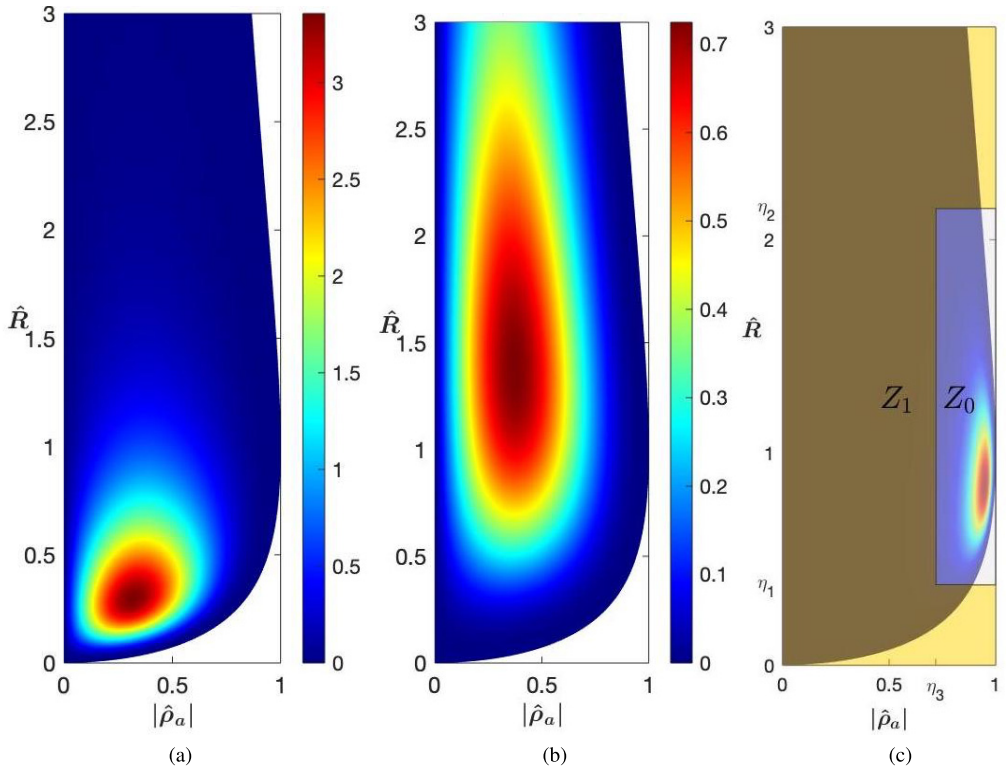


Fig. 2. Joint pdf of $|\hat{\rho}_a|$ and $\hat{R}$ for (a) $R = 1/2, |\rho| = 0, N = 5$, (b) $R = 2, |\rho| = 0, N = 5$, and (c) partitions Z_0 and Z_1 for the two-stage detector.

the 2×2 identity matrix. Thus, z can be written as

$$z \stackrel{d}{=} \mathrm{Tr}\left\{ \mathbf{Q}^{1/2}\mathbf{Q}_d\mathbf{Q}^{1/2} \sum_{k=1}^{N} \mathbf{y}_k\mathbf{y}_k^H \right\}. \tag{18}$$

Since the matrix $\mathbf{M} = \mathbf{Q}^{1/2}\mathbf{Q}_d\mathbf{Q}^{1/2}$ is Hermitian, we can write it as $\mathbf{M} = \mathbf{V}\mathbf{D}\mathbf{V}^H$, where $\mathbf{V}$ is a unitary matrix of eigenvectors, and $\mathbf{D}$ is a diagonal matrix with the two real eigenvalues as its diagonal elements, i.e., $\mathbf{D} = \mathrm{diag}(\lambda_1, \lambda_2)$. Thus

$$z \stackrel{d}{=} \mathrm{Tr}\left\{ \mathbf{D} \sum_{k=1}^{N} \mathbf{V}^H\mathbf{y}_k\mathbf{y}_k^H\mathbf{V} \right\} \stackrel{d}{=} \mathrm{Tr}\left\{ \mathbf{D} \sum_{k=1}^{N} \mathbf{y}_k\mathbf{y}_k^H \right\} \tag{19}$$

where the second equality in distribution follows since unitary transformations do not change the statistical distribution of a standard complex Gaussian vector (see [29, p. 62, Corollary 3.2.1.2]). Letting $\mathbf{y}_k = [u_k, v_k]^T$, we can, thus, write the log-likelihood ratio statistic as

$$z \stackrel{d}{=} \lambda_1 \sum_{k=1}^{N} |u_k|^2 + \lambda_2 \sum_{k=1}^{N} |v_k|^2 \tag{20}$$

where u_k and v_k are independent zero-mean circularly symmetric standard complex Gaussian random variables. Also, it is mentioned in [17] that the two eigenvalues of $\mathbf{Q}_d$ have opposite signs, and using this, it can be shown that the two eigenvalues of $\mathbf{M}$ likewise have opposite signs. We will denote these eigenvalues by λ_1 and λ_2 and assume, without loss of generality, that $\lambda_1 < 0$ and $\lambda_2 > 0$. Thus, the log-likelihood ratio statistic z is a linear combination of two independent complex central chi-squared random variables, the pdf of which has a simpler form compared to the linear combination of two dependent chi-squared random variables given in [17]. The general expression for a linear combination of real chi-squared random variables has been derived in [30]. The statistic z is a linear combination of two complex chi-squared random variables, and its pdf and the expression for $\Pr\{z > \eta\}$ have been derived in Appendix III. The detector z given in (20) has $Z_1 = \{z \in \mathbb{R} : z > \eta\}$, where η can be determined from $\Pr\{z > \eta; H_0\} = P_{\mathrm{FA}}$. Unlike the four detectors discussed previously, the detection performance of z does not depend only on R and $|\rho|$ but rather depends directly on the variances σ_f^2, σ_{g0}^2, and σ_{g1}^2, where σ_{g1}^2 and σ_{g0}^2 represent the variances of g under H_1 and H_0, respectively. It has been shown in [17] that the detection performance of this statistic varies with $\sigma_{g1}^2/\sigma_{g0}^2$, under the assumption $\sigma_f^2 = \sigma_{g0}^2$. Thus, a plot similar to the one given in [17] showing the simulated and the theoretical ROC performance of the log-likelihood ratio is shown in Fig. 3, using the new pdf of the likelihood ratio derived in this article.

IV. Symmetric Two-Stage Change Detector

The symmetric two-stage change detector is a modified version of the two-stage change detector of [7]. An important point worth noting is that the value of R depends on the assignment of the two datasets as f and g, that is, if we interchange f and g, R will become $1/R$. Our goal is to detect change irrespective of whether the change is an increase or a decrease in the variance. In other words, for the purpose of change detection, the two cases $R = R_1$ and $R = 1/R_1$ represent exactly the same scenario (where R_1 is any arbitrary

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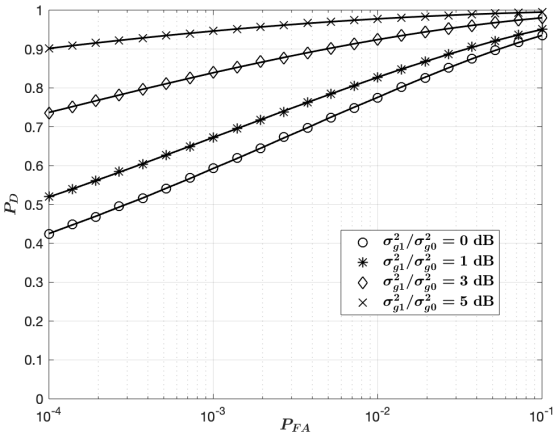


Fig. 3. Simulated (symbols) and theoretical (solid lines) ROC curves for log-likelihood ratio detector for different values of $\sigma^2_{g1}/\sigma^2_{g0}$ with $\rho = 0.7$ under H_0, $\rho = 0$, under H_1, and $N = 5$.

value of the variance ratio). However, $\hat{R}$ is not symmetric in this sense, i.e., it is not invariant to the assignment of images as f and g. Fig. 2(a) and (b) shows the joint pdfs for the 2-D detector for $R = 1/2$ and $R = 2$, respectively, for fixed $N = 5$ and $|\rho| = 0$. It is evident that these two pdfs are not the same and, thus, might give different performances, in spite of the fact that both these cases depict the same scenario in terms of change detection. We can be sure that we do not run into this problem if the pdf of the first stage statistic remains the same even if we invert R. Thus, the symmetric variance ratio detector $\hat{r}$ given by [16] can be used as the first stage, in place of $\hat{R}$. This detector is given by

$$\hat{r} = \min(\hat{R}, 1/\hat{R}). \tag{21}$$

The pdf for $\hat{r}$ has been derived in [16] for the case when $|\rho| = 0$, and this pdf has been used in [17] to determine the thresholds for $\hat{r}$. The complex correlation coefficient between the two pixels acts as a nuisance parameter for the hypothesis test in (4) [31]. However, the formulation in (15) suggests that a nonzero value of $|\rho|$ closer to 1 should be assumed under H_0 in order to compute the threshold for $\hat{r}$. It has been shown in [32] that considering a nonzero value to determine the threshold for $\hat{r}$ in the first stage leads to improved detection performance for the two-stage change detector. Thus, the pdf of $\hat{r}$ for a nonzero coherence $|\rho|$ is given by

$$p_{\hat{r}}(x; R, |\rho|, N) = \frac{\Gamma(2N)}{\Gamma(N)^2} x^{N-1}(1-|\rho|^2)^N \cdot \left(\frac{R^N(x+R)}{[(x+R)^2 - 4xR|\rho|^2]^{N+\frac{1}{2}}} + \frac{R^{-N}(x+R^{-1})}{[(x+R^{-1})^2 - 4xR^{-1}|\rho|^2]^{N+\frac{1}{2}}} \right) \tag{22}$$

for $0 \le x \le 1$ and zero elsewhere. The derivation for this pdf is given in Appendix I. Looking at the expression for the pdf of $\hat{r}$, we can easily see that $\hat{r}$ has the same pdf for $R = R_1$ and $R = R_1^{-1}$ for fixed $|\rho|$ and N, and this makes the detector invariant to image assignment as f and g. Furthermore, since $\hat{r}$ takes values only between zero and one, using $\hat{r}$ instead of $\hat{R}$ for the first stage reduces the decision space for the two-stage detector from $(0 \le |\hat{\rho}_a| < 1, 0 \le \hat{R} < \infty)$ to $(0 \le |\hat{\rho}_a| < 1, 0 \le \hat{r} \le 1)$, i.e., we now need to deal with only a finite range for $\hat{r}$. The joint pdf of $\hat{r}$ and $|\hat{\rho}_a|$ is derived in Appendix II and has a simple form given by

$$p_{|\hat{\rho}_a|,\hat{r}}(x, y; R, |\rho|, N) = p_{|\hat{\rho}_a|,\hat{R}}(x, y; R, |\rho|, N) + p_{|\hat{\rho}_a|,\hat{R}}\left(x, y; \frac{1}{R}, |\rho|, N\right) \tag{23}$$

for $0 \le y \le 1$, $0 \le x \le [4y/(y+1)^2]^{1/2}$, and the pdf is zero elsewhere. The joint pdf for $|\hat{\rho}_a|$ and $\hat{r}$ is plotted in Fig. 4(a) and (b) for the two types of typical H_1 scenarios. Note that Z_1 for the single stage detector $\hat{r}$ has a similar form as that of $|\hat{\rho}_c|$ or $|\hat{\rho}_a|$, i.e., $\hat{r}$ decides that a change has occurred if $\hat{r} \le \eta_1$ (i.e., $\hat{r}$ deviates from unity, but it can deviate only in one direction now as opposed to $\hat{R}$) for some thresold $\eta_1 \in (0, 1)$. Fig. 4(c) shows the joint pdf under an H_0 scenario, along with the partitions Z_0 and Z_1 for this symmetric two-stage detector, which are given by $Z_0 = \{(|\hat{\rho}_a|, \hat{r}) \in [0, 1) \times (0, 1] : \hat{r} > \eta_1 \text{ and } |\hat{\rho}_a| > \eta_2\}$. Thus, we are now required to compute just two thresholds instead of three, and both of these thresholds lie in a finite interval, which simplifies the analysis of the two-stage detector to a great extent. Another statistic that can be used in place of $\hat{R}$ that limits the value between zero to one is given by $\hat{\sigma}_f^2/(\hat{\sigma}_f^2 + \hat{\sigma}_g^2)$, which is a statistic used by Sikaneta and Larson [8].

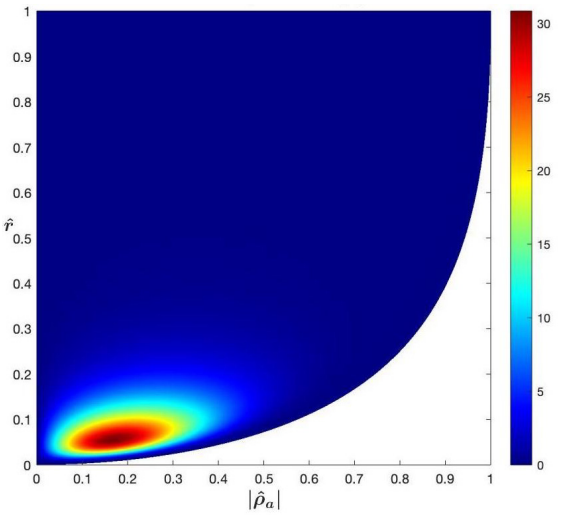
V. Generating ROC Curves for Symmetric Two-Stage Detector

As mentioned in Section II, the first step in generating ROC curves for a detector is determining the region Z_1, more specifically, the required thresholds. The three 1-D detectors $\hat{r}$, $|\hat{\rho}_c|$, and $|\hat{\rho}_a|$ have an identical form for Z_1, which is given by a single threshold, and we can compute that threshold η by substituting $Z_1 \equiv (0, \eta)$ in (2) as follows:

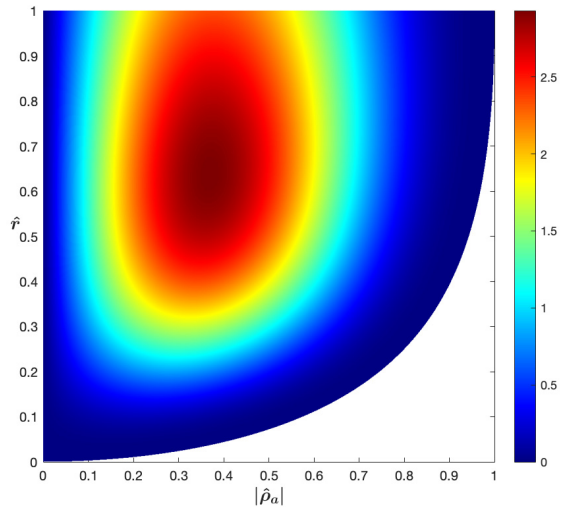
$$P_{\text{FA}} = \Pr\{T \le \eta; H_0\} = \int_0^{\eta} p(T; H_0) dT \tag{24}$$

where T can represent any of these three 1-D statistics. The term on the right-hand side of the above equation is the cdf of the test statistic, and thus, the threshold η can be uniquely determined from a given value of P_{FA}. However, the region Z_1 for the symmetric two stage detector does not have such a simple form. There can exist an infinite number of threshold pairs (η_1, η_2) such that, when we integrate the joint pdf of the 2-D test statistic (under H_0) over Z_1, we get P_{FA}. In other words, a given value of P_{FA} does not yield a unique region Z_1. We require some additional information apart from the given P_{FA} constraint in order to determine η_1 and η_2 uniquely. In order to develop a method for threshold selection, we further partition Z_1 into two sets, Z_{11} $(0 \le \hat{r} \le \eta_1, 0 \le |\hat{\rho}_a| \le 1)$ and Z_{12} $(\eta_1 \le \hat{r} \le 1, 0 \le |\hat{\rho}_a| \le \eta_2)$, as shown in Fig. 5. At this point, we introduce a new parameter $\alpha \in [0, 1]$, which denotes

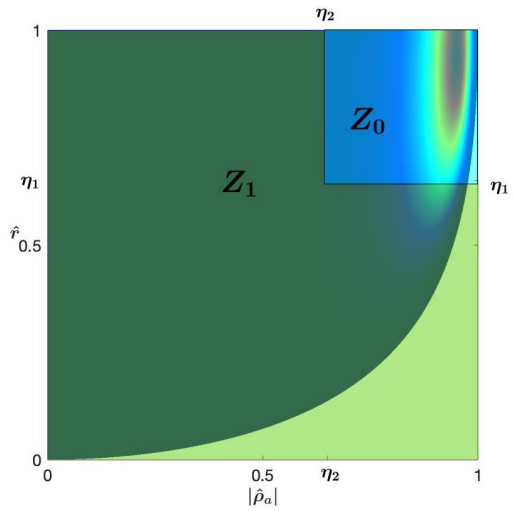
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(a)



(b)



(c)

Fig. 4. Joint pdf of $|\hat{\rho}_a|$ and $\hat{r}$ for (a) $R = 10$, $|\rho| = 0$, $N = 5$, (b) $R = 1$, $|\rho| = 0$, $N = 5$, and (c) $R = 1$, $|\rho| = 0.9$, $N = 5$ with partitions Z_0 and Z_1 for the symmetric two-stage detector.

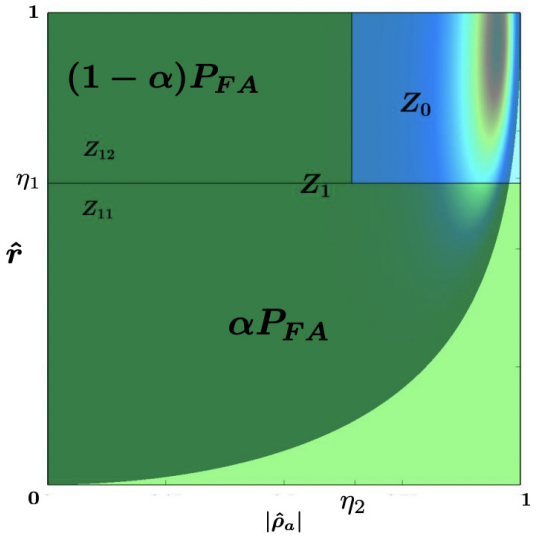


Fig. 5. Distribution of total P_{FA} among the decision space.

the fraction of the total P_{FA}, which is due to Z_{11}. In other words, if we integrate $p(|\hat{\rho}_a|, \hat{r}; H_0)$ over Z_{11}, we get αP_{FA}

$$\begin{aligned}\alpha P_{\text{FA}} &= \int_{Z_{11}} p_{|\hat{\rho}_a|,\hat{r}}(x, y; H_0)dxdy \\ &= \int_0^{\eta_1}\int_0^1 p_{|\hat{\rho}_a|,\hat{r}}(x, y; H_0)dxdy \\ &= \int_0^{\eta_1} p_{\hat{r}}(y; H_0)dy = F_{\hat{r}}(\eta_1; H_0) \end{aligned} \tag{25}$$

where $F_{\hat{r}}(\eta_1)$ denotes the cdf of $\hat{r}$, as given in (37). Since we have the closed-form expression for $F_{\hat{r}}(\eta_1; H_0)$, we can find η_1, which is the value of $\hat{r}$ for which $F_{\hat{r}}(\eta_1; H_0) = \alpha P_{\text{FA}}$. This is equivalent to finding the threshold for the first stage detector $\hat{r}$ with the fixed probability of false alarm constraint equal to αP_{FA}, as discussed earlier in this section. Note that, when we fix α, η_1 is uniquely determined for the given P_{FA} using (25). Since η_1 is now known, we go back to the original P_{FA} constraint in order to find η_2 as follows:

$$\begin{aligned} P_{\text{FA}} &= 1 - \iint_{Z_0} p_{|\hat{\rho}_a|,\hat{r}}(x, y; H_0)dxdy \\ &= 1 - \int_{\eta_1}^1\int_{\eta_2}^1 p_{|\hat{\rho}_a|,\hat{r}}(x, y; H_0)dxdy. \end{aligned} \tag{26}$$

The only unknown quantity in the above equation is η_2, and it is found by computing the zero of the function

$$h(\eta_2) \triangleq \int_{\eta_1}^1\int_{\eta_2}^1 p_{|\hat{\rho}_a|,\hat{r}}(x, y)dxdy - (1 - P_{\text{FA}}) \tag{27}$$

by the Newton–Raphson method, where η_2 is the running variable, and the double-integral is computed numerically in MATLAB for each η_2. Once these thresholds are computed, we have the region Z_1, and thus, we can compute P_{D} using (3). Fig. 6 shows simulated and theoretical ROC curves for the two-stage detector generated using the method

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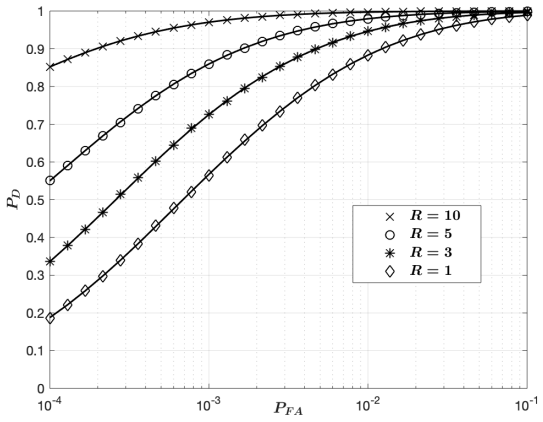


Fig. 6. Simulated (symbols) and theoretical (solid lines) ROC curves for two-stage detector for different values of R under H_1 with $R = 1$ and $|\rho| = 0.9$ under H_0, $|\rho| = 0$ under H_1, $\alpha = 0.3$, and $N = 5$.

described above, for $R = 1$, $|\rho| = 0.9$ under H_0, and $|\rho| = 0$ under H_1 for different values of R under H_1. The value of α used to compute the thresholds is fixed to be 0.3. We can see that the detection performance improves as the value of R under H_1 deviates from unity.

To summarize, for the two-stage detector, there exist an infinite number of threshold pairs (η_1, η_2) that satisfy a given P_{FA} constraint. Thus, in addition to the original P_{FA} constraint, we impose a new constraint on how the thresholds must be selected. This constraint describes the distribution of the total P_{FA} among the entire region Z_1 and is given by α, which represents the contribution of Z_{11} to the total P_{FA}. Thus, fixing a value of α corresponds to selecting an (η_1, η_2) pair, so we fix an α and proceed to determine η_1 and η_2 stepwise. Since each distinct value of α yields a different (η_1, η_2) pair (i.e., a different Z_1), the detection performance will vary as we vary the value of α used for computing thresholds. Our goal now is to find out the value of α such that the thresholds computed using that value give a higher P_{D} compared to other α values. This dependence of the detection performance on α is explored in detail in Section VI.

VI. Behavior of Symmetric Two-Stage Detector for Different Values of α

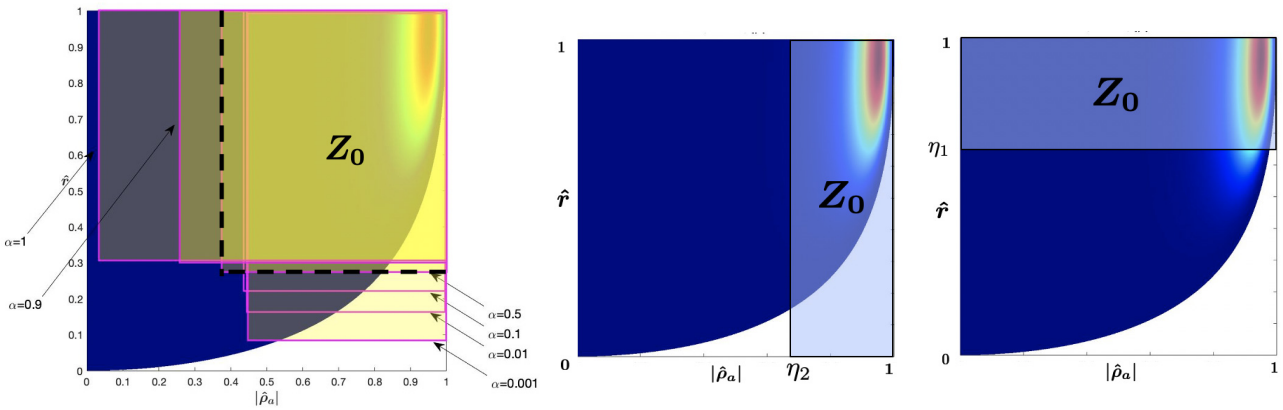
As mentioned earlier, α represents the fraction of the total P_{FA}, which is allotted to Z_{11}. In other words, when the joint pdf of the two-stage detector under H_0 is integrated over Z_{11}, we get αP_{FA}. It is evident from Fig. 5 that, as we decrease α for a fixed P_{FA}, η_1 will also decrease. However, since the entire region Z_1 should integrate to P_{FA}, η_2 will increase as η_1 decreases. As α is made smaller, the rectangle representing the Z_0 region elongates in the vertical direction, as shown in Fig. 7(a). As α approaches zero, η_2 stops changing and attains a constant value. Note that $\alpha = 0$ means that the contribution of Z_{11} to the total P_{FA} is zero, and thus, $\eta_1 = 0$. This effectively means that all the values of $\hat{r}$ are greater than η_1, and the two-stage detector is only comparing the value of $|\hat{\rho}_a|$ with η_2. Thus, α taking the value zero essentially reduces the two-stage detector to the single-stage detector $|\hat{\rho}_a|$. Similarly, as α becomes equal to one, the two-stage detector reduces to $\hat{r}$. These two cases where α takes the extreme values of 0 and 1 are depicted in Fig. 7(b) and (c), respectively. Therefore, as α goes from zero to one, the two-stage detector transitions from $|\hat{\rho}_a|$ to $\hat{r}$ while behaving as a combination of the two detectors for the intermediate values between 0 and 1. An important point to note is that we can use the two-stage detector with $\alpha = 0$ to generate the theoretical ROC curves for $|\hat{\rho}_a|$ for any value of R, which is not possible using the pdf given in (14), since it can only be used for $R = 1$.

We now analyze the dependence of the detection performance on α. Fig. 8 contains the plot of P_{D} as a function of α for $P_{\text{FA}} = 10^{-3}$ and 10^{-4} for $N = 5$. Both figures show the plot of P_{D} versus α for three different values of R under H_1, where $R = 1$ and $|\rho| = 0.9$ under H_0 and $|\rho| = 0$ under H_1. In both figures, if we consider the $R = 1$ curve, the P_{D} is the highest when $\alpha = 0$, i.e., when the two-stage detector is behaving as $|\hat{\rho}_a|$. Furthermore, compared to this highest P_{D}, the P_{D} is significantly lower when the detector has been reduced to $\hat{r}$ ($\alpha = 1$). Recall that it was shown in Fig. 1 that $|\hat{\rho}_c|$ and $|\hat{\rho}_a|$ give the same performance when $R = 1$ under H_1. Thus, we can conclude that, when the change is due to $|\rho|$ only, the two-stage detector with $\alpha = 0$ (i.e., $|\hat{\rho}_a|$) and $|\hat{\rho}_c|$ gives the optimal performance, which is better than any other value of α for the two-stage detector and significantly better than $\hat{r}$. Next, we consider the two curves when the change is due to both R and $|\rho|$, i.e., the curves representing $R = 5$ and $R = 10$ under H_1. In these two cases, the optimal value of α (i.e., the value of $\alpha \in [0, 1]$ achieving the maximum P_{D} for a given P_{FA}) given by α_{opt} is not zero, as indicated in Fig. 8; it is an arbitrary value between zero and one. In general, the value of α_{opt} increases as the value of R under H_1 deviates further from unity, for fixed $|\rho|$, N, and P_{FA}.

The analysis presented so far just reiterates our initial understanding of the SVR and the coherence estimator statistics. Earlier, it was mentioned that the SVR does a better job at detecting large-scale changes, whereas the coherence estimators perform better at detecting subtle changes. This is exactly what we have found with our analysis, i.e., when the change is due to $|\rho|$ only (small-scale change), the optimal detector is $|\hat{\rho}_a|$ (achieved by $\alpha = 0$), whereas, if the change is given by a high value of R under H_1, the optimal detector is $\hat{r}$. For intermediate values of R, α_{opt} is between 0 and 1, which suggests using a combination of $|\hat{\rho}_a|$ and $\hat{r}$. Thus, if the value of R causing the change was available, we could have used this analysis to choose the value of α, which can provide the highest P_{D} for the two-stage change detector.

Although helpful for analysis, this information does not yet help in designing a practical detector, which can achieve improved performance. This is because, in a practical situation, the value of R will not be available to us, and thus, we cannot use this information to find the value of α_{opt}. Thus, in order to find a detector that performs well for all the cases, we need to find a single value of α, which can provide consistently high P_{D} for any value of R under H_1. Let us consider the possible choices, the first one being $\alpha = 1$. If we use the detector with $\alpha = 1$, it would work well for values of R significantly greater than or less than 1. However, if the actual value of R was to be equal to one, as mentioned earlier, we can see from

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(a) (b) (c)

Fig. 7. (a) Transition of the region Z_0 as α varies from 0 to 1. (b) Two-stage detector reducing to $|\hat{\rho}_a|$ when $\alpha = 0$. (c) Two-stage detector reducing to $\hat{r}$ when $\alpha = 1$.

the $R = 1$ curve in Fig. 8 that the decrease in P_{D} by using $\alpha = 1$ when $\alpha_{\mathrm{opt}} = 0$ is significant. Thus, $\alpha = 1$ (i.e., $\hat{r}$) is not a good choice because it gives a significantly low P_{D} when $R = 1$ under H_1. Now, notice that, for the two curves where $R \neq 1$, even though α_{opt} has a fixed nonzero value, choosing any value of α will not give such a steep decrease in P_{D} compared to the highest P_{D} achieved by α_{opt}. Also, we know that $\alpha_{\mathrm{opt}} = 0$ when the change is due to $|\rho|$ only. Thus, $\alpha = 0$ seems a good choice for our detector overall because it is the optimal choice for $R = 1$ under H_1, and even though it is not the optimal value for $R \neq 1$, the P_{D} achieved by $\alpha = 0$ is only slightly less than that achieved by α_{opt}. Furthermore, especially in Fig. 8(b), we can see that there is a decrease in P_{D} as α approaches 0 for the two $R \neq 1$ curves. We can avoid this decrease by choosing a slightly higher value of α, say between 0.1 and 0.2, as long as we do not enter the region where P_{D} starts falling significantly for $R = 1$. Thus, overall, choosing $\alpha \in [0.1, 0.2]$ can give overall better detection performance for the two-stage detector. However, if the slight decrease in P_{D} is not an issue, $|\hat{\rho}_a|$ ($\alpha = 0$) can prove to be a better choice because of its simple structure as it requires computing only a single threshold.

VII. Comparison of ROC Performances of Different Change Detectors

In Section VI, we have presented an argument that, out of all possible choices of α (which includes the two detectors $\hat{r}$ and $|\hat{\rho}_a|$), choosing $\alpha \in [0.1, 0.2]$ gives improved overall detection performance for the two-stage detector. The next question that needs to be explicitly addressed now is whether the two-stage detector provides improved detection performance compared to the classical detector $|\hat{\rho}_c|$. Fig. 9 addresses this question in the form of a plot showing the increase in P_{D} achieved by the two-stage detector with $\alpha = 0.1$ compared to $|\hat{\rho}_c|$ for $N = 5$ and $P_{\mathrm{FA}} = 0.001$. The axes denote the value of R and $|\rho|$ under H_1, and $R = 1$ and $|\rho| = 0.9$ are assumed fixed under H_0 to determine the thresholds. Every point on the grid corresponds to a pair of $(|\rho|, R)$ under H_1, and for this pair, P_{D} is computed for $|\hat{\rho}_c|$ and is subtracted from the P_{D} computed for the two-stage detector with $\alpha = 0.1$. We can see that no point on the grid has a negative value, which means that the two-stage detector with $\alpha = 0.1$ does not attain a P_{D},

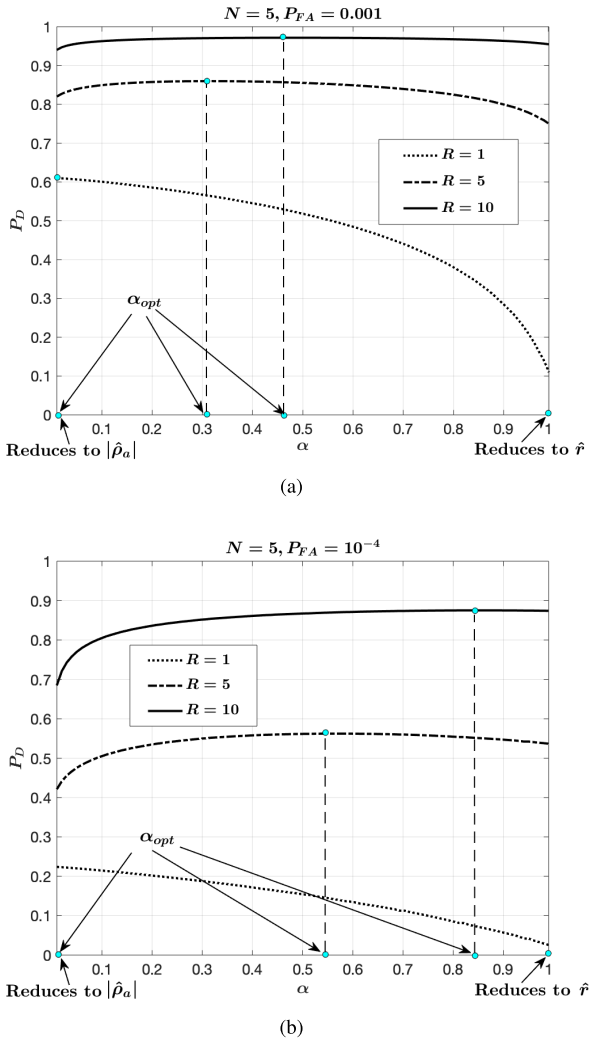


Fig. 8. Plot of α versus P_{D} for $N = 5$ with (a) $P_{\mathrm{FA}} = 10^{-3}$ and (b) $P_{\mathrm{FA}} = 10^{-4}$.

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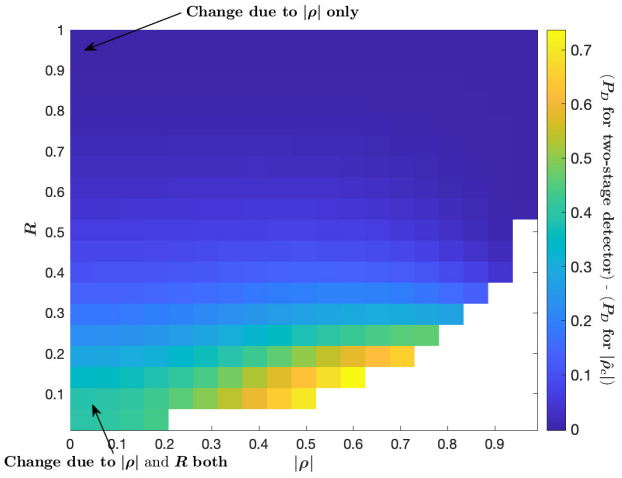


Fig. 9. Difference between the P_{D} achieved by the two-stage detector and $|\hat{\rho}_c|$ for fixed $N = 5$ and $P_{\mathrm{FA}} = 0.001$.

which is less than that attained by $|\hat{\rho}_c|$ for any R and $|\rho|$ under H_1. Furthermore, the bottom-left patch that has $|\rho| \approx 0$ and $R \ll 1$ represents the scenario when the change is due to $|\rho|$ and R both. We can see that, in this region, the increase in P_{D} achieved by the two-stage detector is considerably high (around 0.4). Even though there is a higher improvement for values of $|\rho|$ between 0.4 and 0.7, these values are unlikely to occur under H_1. In any case, the improvement achieved when $|\rho| \approx 0$ is significant. The significance of Fig. 9 is that, even though it only shows values of $R \leq 1$, we are also accounting for the cases $R > 1$ as a result of using the symmetric detector. This means that Fig. 9 includes every type of change that can possibly occur. Thus, the two-stage detector with $\alpha = 0.1$ gives an approximately similar performance to $|\hat{\rho}_c|$ when $R = 1$ under H_1 and a significant improvement in performance over $|\hat{\rho}_c|$ when $R \ll 1$ or $R \gg 1$ under H_1.

Finally, in Fig. 10, we present the theoretical ROC performance comparison for all the detectors mentioned above, i.e., $\hat{r}$, $|\hat{\rho}_a|$, $|\hat{\rho}_c|$, two-stage with $\alpha = 0.1$, and the log-likelihood ratio. The values of R and $|\rho|$ under H_0, which are used to determine the thresholds, are assumed to be 1 and 0.9, respectively, with $N = 5$. $|\rho| = 0$ is assumed for all the three figures under H_1, and the value of R under H_1 is varied. Note that the log-likelihood ratio depends directly on the variances, and these values have been mentioned in the figure, with the subscript indicating the hypothesis. Since the pdf of $|\hat{\rho}_c|$ is invariant to the value of R, we get the same ROC curve for $|\hat{\rho}_c|$ in all three figures. The ROC curves for $|\hat{\rho}_a|$ are generated by using $\alpha = 0.001$ (since the thresholds are computed numerically, we avoid using $\alpha = 0$ exactly to avoid computational issues) in the two-stage detector. Fig. 10(a) depicts the scenario when the change is due to $|\rho|$ only, i.e., $R = 1$ under H_1. It was shown in Fig. 1 that $|\hat{\rho}_c|$ and $|\hat{\rho}_a|$ have the same performance under this scenario; hence, a single curve is plotted for the two statistics in Fig. 10(a). As discussed in Section VI, P_{D} achieved by the two-stage detector with $\alpha = 0.1$ is almost equal to the optimal value achieved by $|\hat{\rho}_c|$ (or $|\hat{\rho}_a|$, i.e., two-stage with $\alpha = 0$). Since there is no change

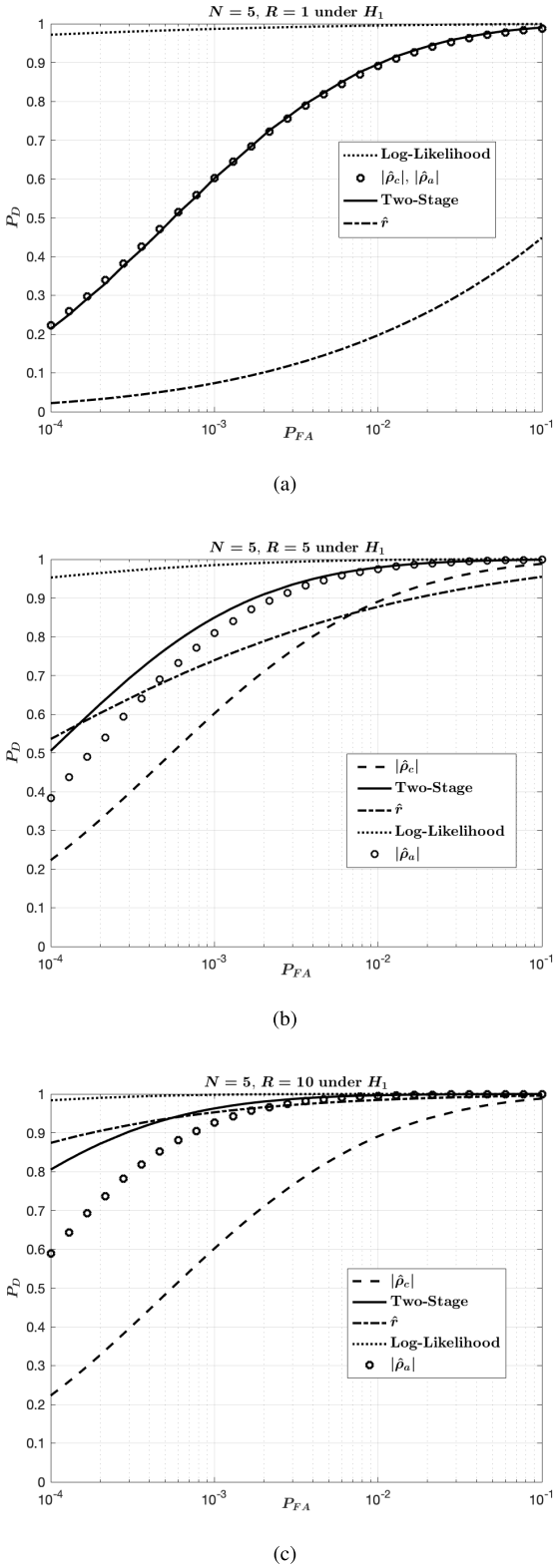


Fig. 10. ROC curves for the five detectors for $|\rho| = 0.9$ under H_0, $\rho = 0$ under H_1, and $\sigma_f = \sigma_{g0} = 1$, i.e., $R = 1$ under H_0. (a) $\sigma_{g1} = 1$, i.e., $R = 1$ under H_1. (b) $\sigma_{g1} = \sqrt{0.2}$, i.e., $R = 5$ under H_1. (c) $\sigma_{g1} = \sqrt{0.1}$, i.e., $R = 10$ under H_1.

due to R, $\hat{r}$ performs poorly. Thus, in this case, we see that, when the change is due to $|\rho|$ only, $|\hat{\rho}_c|$, $|\hat{\rho}_a|$, and two-stage detector with $\alpha = 0.1$ perform better than $\hat{r}$, which performs

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poorly as it fails to detect the change due to $|\rho|$. In Fig. 10(b), the change is due to R as well, with $R = 5$ under H_1, and as a result, we can see that the performance of $\hat{r}$ has improved compared to the case when $R = 1$ under H_1. We can also see that the two-stage detector gives a better performance than the other three detectors, and $|\hat{\rho}_a|$ also approximately achieves this same performance. Note that, in this case, not only the performance of $|\hat{\rho}_a|$ is different from that of $|\hat{\rho}_c|$ but also considerably better since $|\hat{\rho}_c|$, being independent of R, fails to capture the changes due to R. Finally, in Fig. 10(c), for lower values of P_{FA}, $\hat{r}$ performs slightly better, and $|\hat{\rho}_a|$ performs poorer compared to the two-stage detector. However, we can see that, for values of $P_{\text{FA}} \geq 0.001$, these three detectors give an almost equal performance. In all the cases, the log-likelihood ratio shows the highest achievable P_{D}.

To summarize, we can state two important observations from analyzing Fig. 10. The first observation involves comparing the performance of the two-stage detector with the two popularly used methods, $|\hat{\rho}_c|$ and $\hat{r}$. We can observe that, among these two detectors, $|\hat{\rho}_c|$ gives a better performance when the change is due to $|\rho|$ only, and $\hat{r}$ gives a better performance when the change is due to R and $|\rho|$ both. However, the two-stage detector matches the performance of these two detectors and proves to be a single detector, which consistently gives a high detection performance, especially when one of these two detectors fails. The second case is regarding the curious case of $|\hat{\rho}_a|$, which achieves a performance, which is only slightly poorer than the two-stage detector. It was mentioned in Section II-B that $|\hat{\rho}_a|$ provides accurate estimates of $|\rho|$ only when $R \approx 1$. However, Fig. 10 shows that $|\hat{\rho}_a|$ is achieving a better performance compared to $|\hat{\rho}_c|$ when $R \neq 1$ under H_1. The reason for this, as explained in [7], is that $|\hat{\rho}_a|$ and $|\hat{\rho}_c|$ contain the arithmetic and the geometric means of the sample variances in their denominators. As a result, $|\hat{\rho}_a| \leq |\hat{\rho}_c|$ always, and $|\hat{\rho}_a|$ decreases as the sample variances move further apart from each other. Thus, it becomes easier for $|\hat{\rho}_a|$ to detect change because it is further from unity, especially when $R \neq 1$. In summary, out of the four realizable detectors considered in this article, the two-stage detector with $\alpha = 0.1$ achieves higher overall detection performance compared to the other three detectors, whereas $|\hat{\rho}_a|$ proves to be a simpler alternative since it achieves P_{D} only slightly less than the two-stage detector. It was observed that the behavior of detection performance of the various detectors described here holds true and is useful for values up to $N \approx 12$. However, as N further increases and takes larger values, such as 16 and 25, all the detectors start achieving an equal performance of $P_{\text{D}} = 1$ for the range of false alarm probabilities considered here.

VIII. Experimental Results

In this section, the four change detectors $\hat{r}$, $|\hat{\rho}_c|$, $|\hat{\rho}_a|$, and the two-stage detector with $\alpha = 0.1$ are implemented on a SAR dataset, made available publicly by the Air Force Research Laboratory, as described in [34]. Fig. 11 shows the magnitude image of one of the two SAR images among which the change is to be detected. Generally, the two SAR images taken at different times are used to produce the value of the raw



Fig. 11. Magnitude of the reference SAR image from the dataset given in [34].

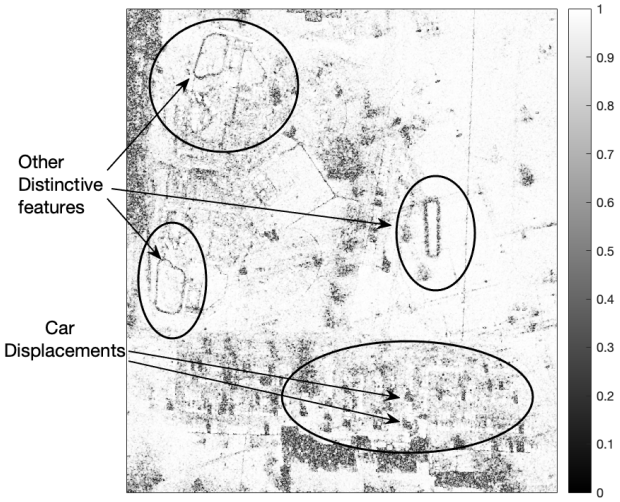


Fig. 12. Raw statistic $|\hat{\rho}_c|$ for the AFRL dataset given in [34] giving a gray-scale image where the change is represented by a lower value of coherence, i.e., a darker pixel.

(prior to thresholding) statistic for each pixel, and plotting this value of the raw statistic for each pixel results in a gray-scale change image. Such a change image for the data from [34] is shown in Fig. 12, which contains the value of $|\hat{\rho}_c|$ for each pixel. Since, throughout this article, binary thresholding algorithms have been discussed, in this section, we explore the binary thresholding capability of the four change detectors on the SAR dataset, i.e., applying the detectors to the two SAR images must result in a binary (black and white) image, where a black pixel denotes change and a white pixel denotes no change. Fig. 12 is a gray-scale image that is used as a reference to determine areas of change. Some areas of change have been described in [34], which are indicated in Fig. 12. Small clusters of dark pixels shown in the elliptical region correspond to car displacements. Apart from this, other distinctive features in Fig. 12 can also be considered as reference regions, as shown in the figure. An important point worth mentioning is that Fig. 12 is not the ground truth for binary

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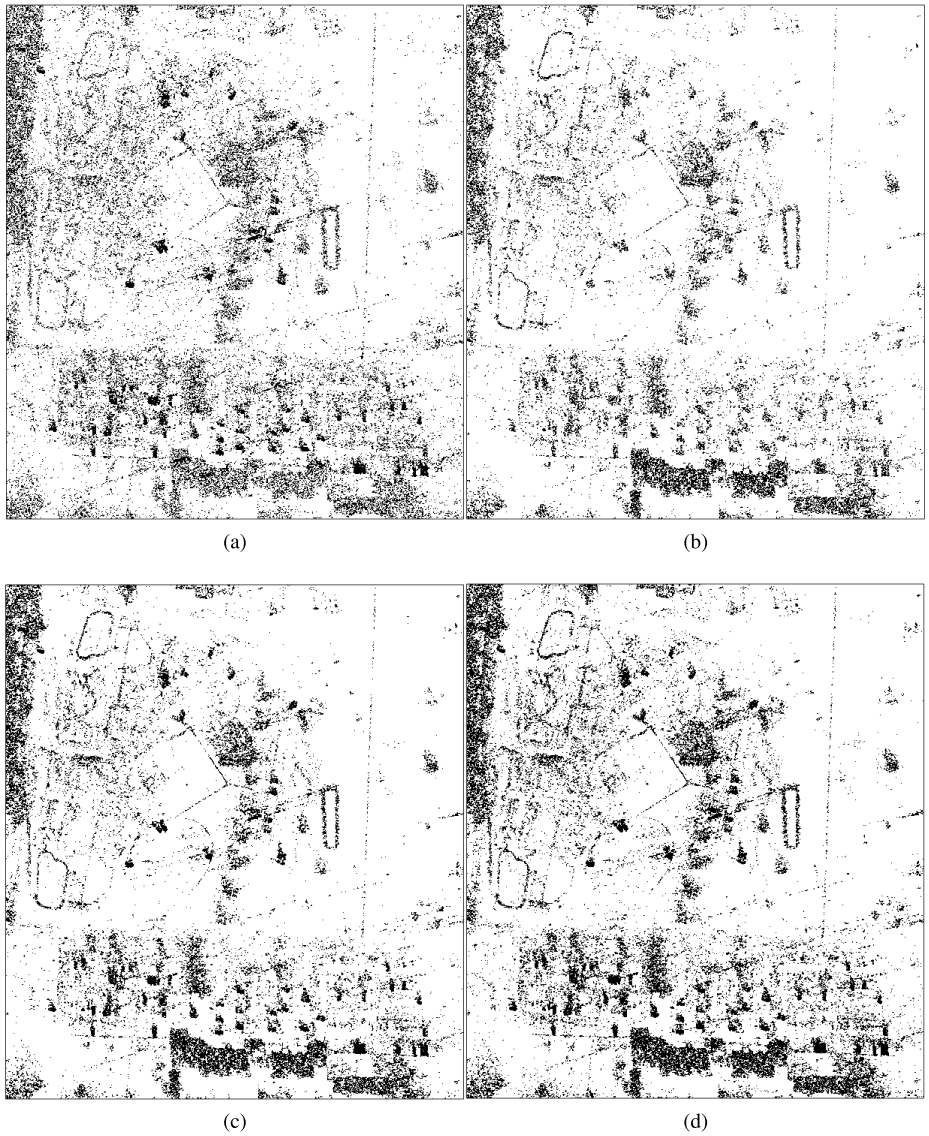


Fig. 13. All the four detectors implemented on AFRL dataset given in [34]. (a) SSVR $\hat{r}$. (b) Classical coherence estimator $|\hat{\rho}_c|$. (c) Alternative coherence estimator $|\hat{\rho}_a|$. (d) Two-stage detector with $\alpha = 0.1$.

change detection; it is a gray-scale image that is used as a reference to allow us to form a subjective view regarding which regions can be considered as a change. The analysis of the algorithms presented in this section is an attempt to find which algorithms can produce change images that put more emphasis on changes of interest and suppress uninteresting changes. The terms "changes of interest" and "uninteresting changes" can be subjective; however, in this study, they are used in the context similar to the one given in [35], where uninteresting changes refer to clutter regions (regions with low reflectivity, shadows, and vegetation), and changes of interest are man-made changes that we are particularly looking for like vehicle tracks or the displacement of the parked cars.

The thresholds for each of the detectors are computed using $R = 1$ and $|\rho| = 0.9$, a 3×4 rectangular window giving $N = 12$ is considered, and the probability of false alarm is 0.001. This choice of window size has been used in this article based on a comparison with images generated using window sizes of 3×3 and 4×4. It was seen that a 3×4 window is the smallest window for which the changes in the images are clearly visible for all four detectors. More generally, the effective number of looks (ENL) must be estimated, which gives the number of independent pixels, and the value of N must be chosen accordingly. However, it was observed that taking the window size $N = 12$ and using that for computing the thresholds generated change images that were clearly displaying the changes between the two images, and thus, these change images were used. Also, ideally, the value of $|\rho|$ should be estimated from the data, but this can only be done if we have samples available with known ground truth showing no change. However, the pixelwise ground truth for the particular SAR dataset that was used was not available; thus, it was not possible to know which pixels have undergone change and which have not. Typically, the value is greater than 0.4, and the specific value of 0.9 was motivated from [7], where this value was used for the same SAR dataset in order to generate the change images. Fig. 13 shows the binary change images produced by the four detectors. In some

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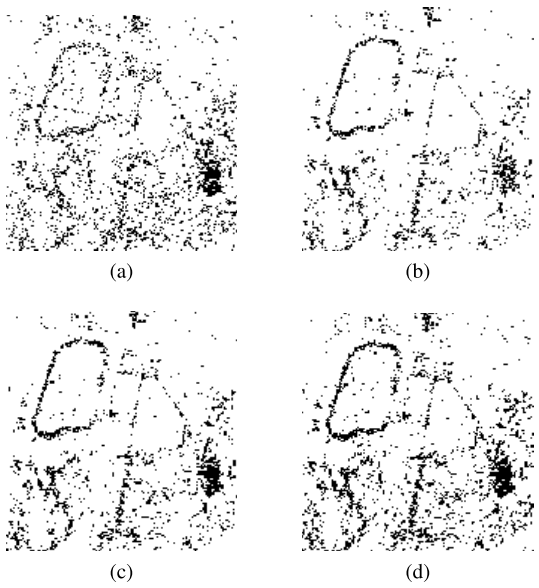


Fig. 14. Close-up view of the region indicated in the top-left corner of Fig. 12, generated by (a) $\hat{r}$, (b) $|\hat{\rho}_c|$, (c) $|\hat{\rho}_a|$, and (d) two-stage detector with $\alpha = 0.1$.

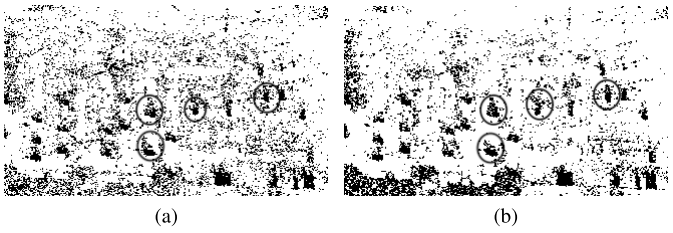


Fig. 15. Close-up view of the encircled car displacements as indicated in Fig. 12, generated by (a) $\hat{r}$ and (b) $|\hat{\rho}_a|$.

regions, $\hat{r}$ detects the change, which is not detected by the other three detectors, which can be seen in the form of a greater number of black pixels in Fig. 13(a), which are not present in the other three figures. Fig. 14 illustrates this with the help of a zoomed-in image of the region indicated in the top-left corner of Fig. 12, where we can see that $\hat{r}$ certainly gives a greater number of black pixels compared to the $|\hat{\rho}_c|$ and $|\hat{\rho}_a|$, while this difference is not that significant compared to the two-stage detector. It is possible that these may be false alarms caused by uninteresting changes arising in clutter regions as explained earlier. We can see in Fig. 13 that these large-scale car displacements are better captured by $\hat{r}$, $|\hat{\rho}_a|$, and the two-staged detector in the form of densely packed clusters having a greater number of black pixels compared to $|\hat{\rho}_c|$. However, $\hat{r}$ fails to suppress the regions surrounding the car displacements, and we can see a larger number of black pixels surrounding the cars, as mentioned earlier. Finally, $|\hat{\rho}_a|$ and the two-stage detector can be seen to improve upon both these aspects with clearly visible car displacements and a smaller number of black pixels in the surrounding regions compared to $\hat{r}$. The difference between $|\hat{\rho}_a|$ and $\hat{r}$ has been illustrated in Fig. 15 by focusing on the car displacements.

It was shown in Section VII that $|\hat{\rho}_a|$ and the two-stage detector give approximately the same performance. Even here, we can see that $|\hat{\rho}_a|$ and the two-stage detector produce almost identical change images, where the two-stage detector gives a slightly greater number of false alarms. The changes are more easily visible in the change images produced by these two detectors compared to the ones produced by $|\hat{\rho}_c|$ and $\hat{r}$, as discussed earlier. Overall, the binary change images produced by $|\hat{\rho}_a|$ and the two-stage detector capture a significant fraction of the features present in the gray-scale image obtained in Fig. 12. Thus, we can use this compressed representation without missing out on a significant number of changes, and this can prove advantageous in terms of data compression, especially if one is dealing with a large number of images. Furthermore, another way of generating change images can be considered as follows: binary thresholding by the two-stage change detector as considered above is used to indicate the pixels having undergone change, i.e., we obtain a binary image showing the pixels that have changed as black pixels and those that have not as white. Next, gray-scale change detection by any suitable change detector (by plotting the actual value of the statistic, as shown in Fig. 12) can be performed on only those pixels that have been detected as change by the two-stage detector. The result of this process would be an image having white pixels for the areas where the two-stage detector has declared no-change and gray-scaled pixels instead of the black pixels generated by a binary detector that it has declared as change, thus achieving better contrast.

IX. Conclusion

The problem of detecting areas of change between SAR images of the same scene, captured at two different time instants, is explored in this study. Particularly, the two-stage change detector proposed by Cha *et al.* [7] has been studied in detail. This detector is a two-dimensional detector, where the dimensions correspond to the SVR $\hat{R}$, and Berger's alternative coherence estimator $|\hat{\rho}_a|$. It has been shown that replacing the SVR test statistic with the symmetric detector $\hat{r} = \min(\hat{R}, 1/\hat{R})$ yields a simpler form for the two-stage detector. The pdf and the cdf for this symmetric ratio detector for a nonzero correlation coefficient ρ have been derived. The pdf for the symmetric two-stage detector has also been derived. An approach for computing the thresholds with the help of this density function for a fixed probability of false alarm constraint is proposed. It is shown that, in order to uniquely determine the thresholds, the introduction of another parameter α is required. The two-stage detector effectively reduces to the SSVR detector when $\alpha = 1$, whereas, when $\alpha \approx 0$, it reduces to the alternative coherence estimator. α taking a value between zero and one results in the two-stage detector behaving as a combination of the two statistics $|\hat{\rho}_a|$ and $\hat{r}$.

The dependence of the detection performance on α was explored in detail, and it has been shown that the optimal value of α depends on the true value of R and $|\rho|$. However, considering a fixed value of $\alpha \in [0.1, 0.2]$ still achieves a better detection performance compared to the classical coherence estimator $|\hat{\rho}_c|$ and the SSVR $\hat{r}$, for $P_{\text{FA}} \geq 0.001$.

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For lower values of P_{FA} and values of R further away from unity under the change scenario, the performance of $\hat{r}$ can possibly be better than the two-stage detector, but, for values of R close to one, $\hat{r}$ is unable to capture the phase changes. Theoretical ROC curves of all the four detectors were compared with the optimal log-likelihood ratio detector, which demonstrated that the two-stage detector with $\alpha = 0.1$ achieves consistently better performance when one of the two other detectors fails to do so. However, since the probability of detection achieved by $|\hat{\rho}_a|$ is only slightly less than that of the two-stage detector, the alternative coherence estimator can prove to be a better option since it only involves a single threshold. In this article, all the ROC curves were plotted for $0.0001 \leq P_{\text{FA}} \leq 0.1$, and this particular range of P_{FA} was used as per the similar choice made in a number of papers, such as [7] and [17]. One of the directions of future work is to explore the performance of these various detectors for lower values of P_{FA}. All of the four detectors were also implemented on a SAR dataset, and qualitatively better change images were generated by $|\hat{\rho}_a|$ and the two-stage detector compared to $\hat{r}$ and $|\hat{\rho}_c|$. Finally, a simplified expression for the pdf of the log-likelihood statistic was also derived.

Appendix I
Derivation for PDF and CDF of $\hat{r}$

The values of $|\rho|$ and N do not change throughout this derivation; thus, for simplicity, they have been dropped from the notation. For instance, $p_{\hat{R}}(x; R, |\rho|, N)$ will be written as $p_{\hat{R}}(x; R)$. The approach to derive the pdf of $\hat{r}$ is similar to the one used in [16]. It can be seen from (21) that the support of the pdf of $\hat{r}$, $p_{\hat{r}}(x; R)$, is $0 \leq x \leq 1$. Thus, for any $0 \leq a < b \leq 1$, we can write

$$P(a < \hat{r} < b; R) = P(a < \hat{R} < b, \hat{R} \leq 1; R) + P(a < \hat{R}^{-1} < b, \hat{R} > 1; R). \quad (28)$$

Since a and b lie between 0 and 1, the event $\hat{R} \in [a, b]$ is a subset of the event $\hat{R} \leq 1$, and thus

$$P(a < \hat{R} < b, \hat{R} \leq 1; R) = P(a < \hat{R} < b; R). \quad (29)$$

Similarly, $\hat{R}^{-1} \in [a, b]$, which can also be written as $\hat{R} \in [1/b, 1/a]$, is a subset of the event $\hat{R} > 1$. Hence

$$P\left(a < \hat{R}^{-1} < b, \hat{R} > 1; R\right) = P\left(b^{-1} < \hat{R} < a^{-1}; R\right). \quad (30)$$

Thus, using (29) and (30) in (28), we get

$$P(a < \hat{r} < b; R) = P(a < \hat{R} < b; R) + P\left(b^{-1} < \hat{R} < a^{-1}; R\right). \quad (31)$$

We now consider $p_{\hat{R}}(x^{-1}; R^{-1})$, which can be obtained from the expression for $p_{\hat{R}}(x; R)$, which is given in (6)

$$\begin{aligned} p_{\hat{R}}\left(x^{-1}; R^{-1}\right) &= K \cdot \frac{(1-|\rho|^2)^N (x^{-1}+R^{-1}) R^{-N} x^{1-N}}{[(x^{-1}+R^{-1})^2 - 4x^{-1}R^{-1}|\rho|^2]^{N+\frac{1}{2}}} \\ &= K \cdot \frac{(1-|\rho|^2)^N (x+R) R^N x^{N-1}}{[(x+R)^2 - 4xR|\rho|^2]^{N+\frac{1}{2}}} . x^2 \\ p_{\hat{R}}\left(x^{-1}; R^{-1}\right) &= p_{\hat{R}}(x; R) . x^2 \end{aligned} \quad (32)$$

where $K = \Gamma(2N)/\Gamma(N)^2$. We now consider the second term in the right-hand side of (31), and we get

$$\begin{aligned} P\left(b^{-1} < \hat{R} < a^{-1}; R\right) &= \int_{b^{-1}}^{a^{-1}} p_{\hat{R}}(x; R) dx \\ &= \int_a^b p_{\hat{R}}\left(y; \frac{1}{R}\right) dy \end{aligned} \quad (33)$$

by using the substitution $x = 1/y$ and the result in (32). Substituting (33) in (31), we get

$$\int_a^b p_{\hat{r}}(y; R) dy = \int_a^b \left[p_{\hat{R}}(y; R) + p_{\hat{R}}\left(y; \frac{1}{R}\right) \right] dy \quad (34)$$

which gives us the final result

$$p_{\hat{r}}(y; R, |\rho|, N) = p_{\hat{R}}(y; R, |\rho|, N) + p_{\hat{R}}\left(y; \frac{1}{R}, |\rho|, N\right) \quad (35)$$

which can be further expanded and written in the form given in (22). Furthermore, using $a = 0$ and $b = \eta$ in (34), we get

$$\int_0^\eta p_{\hat{r}}(y; R) dy = \int_0^\eta p_{\hat{R}}(y; R) dy + \int_0^\eta p_{\hat{R}}\left(y; \frac{1}{R}\right) dy. \quad (36)$$

The integral on the left-hand side gives the cdf of $\hat{r}$, and the two integrals on the right-hand side are in terms of the cdf of $\hat{R}$, under true parameter values R and R^{-1}. Thus, the cdf of $\hat{r}$ is given by

$$F_{\hat{r}}(\eta; R, |\rho|, N) = F_{\hat{R}}(\eta; R, |\rho|, N) + F_{\hat{R}}(\eta; R^{-1}, |\rho|, N) \quad (37)$$

where $F_{\hat{R}}(\eta; R, |\rho|, N)$ is the cdf of $\hat{R}$ in (7), where

$$\begin{aligned} \lambda_1(\eta) &= (R-\eta) + \sqrt{(R+\eta)^2 - 4\eta|\rho|^2 R} \\ \lambda_2(\eta) &= (R-\eta) - \sqrt{(R+\eta)^2 - 4\eta|\rho|^2 R} \\ l(\eta) &= -\frac{\lambda_2(\eta)}{\lambda_1(\eta)}. \end{aligned}$$

Furthermore, it can be shown that $\lambda_1(\eta) > 0$ as follows:

$$|\rho|^2 < 1 \implies 2|\rho|^2 - 1 < 1 \implies 2R\eta(2|\rho|^2 - 1) < 2R\eta$$

since R and η are nonnegative. Thus

$$R^2 + \eta^2 - 2R\eta(2|\rho|^2 - 1) > R^2 + \eta^2 - 2R\eta = (R-\eta)^2.$$

Also, note that

$$R^2 + \eta^2 - 2R\eta(2|\rho|^2 - 1) = (R+\eta)^2 - 4\eta|\rho|^2 R.$$

Thus, we finally get

$$|R - \eta| < \sqrt{(R+\eta)^2 - 4\eta|\rho|^2 R}.$$

Thus, even if $R - \eta < 0$, $\lambda_1(\eta)$ will always be positive, and we get

$$F_{\hat{R}}(\eta; R, |\rho|, N) = F_G(l(\eta), N). \quad (38)$$

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APPENDIX II
DERIVATION FOR JOINT PDF OF $|\hat{\rho}_a|$ AND $\hat{r}$

In order to derive the expression for $p_{|\hat{\rho}_a|,\hat{r}}(x, y)$, we use a similar procedure as the one used in the derivation of $p_{\hat{r}}(y)$. We begin by considering

$$\begin{aligned} &P(a < \hat{r} < b, c < |\hat{\rho}_a| < d; R) \\ &= P(a < \hat{R} < b, c < |\hat{\rho}_a| < d, \hat{R} \leq 1; R) \\ &\quad + P(a < \hat{R}^{-1} < b, c < |\hat{\rho}_a| < d, \hat{R} > 1; R) \end{aligned} \tag{39}$$

where $0 \leq a < b \leq 1$ and $0 \leq c < d \leq 1$. Similar to (31), we can write the above equation as

$$\begin{aligned} &P(a < \hat{r} < b, c < |\hat{\rho}_a| < d; R) \\ &= P(a < \hat{R} < b, c < |\hat{\rho}_a| < d; R) \\ &\quad + P(b^{-1} < \hat{R} < a^{-1}, c < |\hat{\rho}_a| < d; R). \end{aligned} \tag{40}$$

Considering $p_{|\hat{\rho}_a|,\hat{R}}(x, y^{-1}; R^{-1}, |\rho|, N)$, similar to (32), we can prove that

$$p_{|\hat{\rho}_a|,\hat{R}}(x, y^{-1}; R^{-1}, |\rho|, N) = p_{|\hat{\rho}_a|,\hat{R}}(x, y; R, |\rho|, N).y^2. \tag{41}$$

Considering the second term in the right-hand side of (40)

$$\begin{aligned} &P(b^{-1} < \hat{R} < a^{-1}, c < |\hat{\rho}_a| < d; R) \\ &\quad = \int_c^d \int_{1/b}^{1/a} p_{|\hat{\rho}_a|,\hat{R}}(x, y; R) dy dx \\ &\quad = \int_c^d \int_a^b p_{|\hat{\rho}_a|,\hat{R}}\left(x, z; \frac{1}{R}\right) dz dx \end{aligned} \tag{42}$$

where the substitution $y = 1/z$ and (41) have been used. We can now write (40) as

$$\begin{aligned} &\int_c^d \int_a^b p_{|\hat{\rho}_a|,\hat{r}}(x, z; R) dz dx \\ &\quad = \int_c^d \int_a^b p_{|\hat{\rho}_a|,\hat{R}}(x, z; R) dz dx \\ &\quad\quad + \int_c^d \int_a^b p_{|\hat{\rho}_a|,\hat{R}}\left(x, z; \frac{1}{R}\right) dz dx. \end{aligned} \tag{43}$$

Thus, we get the final result for the expression for the joint pdf of $\hat{r}$ and $|\hat{\rho}_a|$ as

$$\begin{aligned} p_{|\hat{\rho}_a|,\hat{r}}(x, z; R, |\rho|, N) &= p_{|\hat{\rho}_a|,\hat{R}}(x, z; R, |\rho|, N) \\ &\quad + p_{|\hat{\rho}_a|,\hat{R}}\left(x, z; \frac{1}{R}, |\rho|, N\right). \end{aligned} \tag{44}$$

APPENDIX III
DERIVATION FOR PDF OF LOG-LIKELIHOOD RATIO

The random variables $z_1 = \lambda_1 \sum_{k=1}^{N} |u_k|^2$ and $z_2 = \lambda_2 \sum_{k=1}^{N} |v_k|^2$ are negatively and positively scaled independent complex central chi-squared random variables, respectively. In general, the pdf and moment generating function (mgf) pair for such a scaled chi-squared random variable $Y_{\beta,N} = \beta \cdot \chi^2_N$ are given by [36]

$$\begin{aligned} p_{Y_{\beta,N}} = \frac{(y/\beta)^{N-1} e^{-y/\beta}}{|\beta|(N-1)!} &\longleftrightarrow \left(\frac{-1/\beta}{s - \frac{1}{\beta}}\right)^N = \Phi_{Y_{\beta,N}}(s) \\ &= \mathbb{E}\left(e^{sY_{\beta,N}}\right) \end{aligned} \tag{45}$$

where the pdf and mgf are defined for $y < 0$ and $\mathrm{Re}(s) > 1/\beta$, respectively, if $\beta < 0$, and for $y > 0$ and $\mathrm{Re}(s) < 1/\beta$ if $\beta > 0$. The expression for $\Pr\{Y_{\beta,N} \geq y\}$ has been provided in [37] for $\beta = 1$, from which the expression for a general value of β can be derived using the result on scaling of random variables given in [38]

$$\Pr\{Y_{\beta,N} \geq y\} = \begin{cases} \dfrac{\Gamma(N, y/\beta)}{(N-1)!}, & \text{for } y > 0 \\ 1, & \text{for } y \leq 0 \end{cases} \tag{46}$$

when $\beta > 0$. When $\beta < 0$

$$\Pr\{Y_{\beta,N} \geq y\} = \begin{cases} 0, & \text{for } y > 0 \\ 1 - \dfrac{\Gamma(N, y/\beta)}{(N-1)!}, & \text{for } y \leq 0 \end{cases} \tag{47}$$

where $\Gamma(K, x)$ represents the incomplete gamma function given by [24]

$$\Gamma(K, x) = (K-1)! e^{-x} \sum_{i=1}^{K-1} \frac{x^i}{i!} \tag{48}$$

when K is a positive integer. When $x = 0$, this function reduces to the gamma function, i.e., $\Gamma(K, 0) = \Gamma(K) = (K-1)!$. Since z_1 and z_2 are independent, the mgf $\Phi(s)$ for s such that their sum z for $1/\lambda_1 < \mathrm{Re}(s) < 1/\lambda_2$ is given by

$$\Phi(s) = \left(\frac{\frac{1}{\lambda_1}}{s - \frac{1}{\lambda_1}}\right)^N \left(\frac{\frac{1}{\lambda_2}}{s - \frac{1}{\lambda_2}}\right)^N. \tag{49}$$

This mgf has a partial fraction expansion of the form

$$\begin{aligned} \Phi(s) &= \sum_{m=1}^{N} \left[\frac{c_m}{\left(s - \frac{1}{\lambda_1}\right)^m} + \frac{d_m}{\left(s - \frac{1}{\lambda_2}\right)^m} \right] \\ &= \sum_{m=1}^{N} \left[(-\lambda_1)^m c_m \left(\frac{-\frac{1}{\lambda_1}}{s - \frac{1}{\lambda_1}}\right)^m \right. \\ &\quad \left. + (-\lambda_2)^m d_m \left(\frac{-\frac{1}{\lambda_2}}{s - \frac{1}{\lambda_2}}\right)^m \right]. \end{aligned} \tag{50}$$

Finally, from (45) and (50), we can write the pdf of z, given by $p(z)$ as follows:

$$p(z) = \sum_{m=1}^{N} \left[(-\lambda_1)^m c_m p_{Y\lambda_1,m}(z) + (-\lambda_2)^m d_m p_{Y\lambda_2,m}(z) \right] \tag{51}$$

and the expression for $\Pr\{z \geq y\}$ from (46) and (47) as follows:

$$\begin{aligned} \Pr\{z \geq y\} = \sum_{m=1}^{N} \left[(-\lambda_1)^m c_m \left(1 - \frac{\Gamma(m, y/\lambda_1)}{(m-1)!}\right) \right. \\ \left. + (-\lambda_2)^m d_m \right] \end{aligned} \tag{52}$$

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for $y \leq 0$. For $y > 0$

$$\Pr\{z \geq y\} = \sum_{m=1}^{N} \left[(-\lambda_2)^m d_m \frac{\Gamma(m, y/\lambda_2)}{(m-1)!} \right]. \tag{53}$$

The partial fraction coefficients are given by [39]

$$c_m = \frac{1}{(N-m)!} \left[\frac{\partial^{N-m}}{\partial s^{N-m}} \left(\left(s - \frac{1}{\lambda_1} \right)^N \Phi(s) \right) \right]_{s=\frac{1}{\lambda_1}} \tag{54}$$

$$d_m = \frac{1}{(N-m)!} \left[\frac{\partial^{N-m}}{\partial s^{N-m}} \left(\left(s - \frac{1}{\lambda_2} \right)^N \Phi(s) \right) \right]_{s=\frac{1}{\lambda_2}}. \tag{55}$$

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